

CIT305: Social Media Applications

Chatbots & Conversational Interfaces

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The Core Concept: What is a Chatbot?

- **Definition:** An automated program that simulates human conversation through text or voice.
- **Goal:** To interact with users in a natural, conversational manner.
- **Key Function:** Simplifies communication and automates tasks that would typically require human interaction.
- **Why it Matters:** Chatbots enable instant, 24/7 engagement on a massive scale.

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The Shift from Graphical to Conversational

- **Graphical User Interfaces (GUIs):** Rely on buttons, menus, and icons. (e.g., websites, apps). This requires users to navigate a pre-defined path.
- **Conversational User Interfaces (CUIs):** The interface is the conversation itself. Users simply express their intent in natural language.

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A Brief History of Chatbots

- **1960s - ELIZA:** The first conversational program. A simple script that mimicked a psychotherapist.
- **1990s - ALICE:** An open-source chatbot that used XML-based rules. More advanced, but still rule-based.
- **2000s - Siri & Alexa:** The introduction of voice assistants brought conversational AI to the mainstream.
- **2016+ - The Social Media Boom:** Facebook Messenger opens its platform to developers, leading to a massive surge in business chatbots. This is when the modern era of social media chatbots began.

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The Social Media Imperative

- **Audience:** Billions of users are already active on social media platforms.
- **Habit:** Messaging is a primary form of communication. Brands can meet customers where they already are.
- **Data:** Social media platforms provide rich data for personalizing interactions.
- **Seamless Experience:** Users can move from a post to a direct message to a purchase without leaving the app.

Two Main Types of Chatbots

Rule-Based Chatbots:

- **How they work:** Follow a predefined decision tree or set of "if/then" rules.
- **Strengths:** Predictable, easy to build for simple tasks (e.g., FAQs, simple order tracking).
- **Weaknesses:** Cannot handle unscripted or complex queries. The user hits a "dead end."



Two Main Types of Chatbots Contd.

AI-Powered Chatbots:

- **How they work:** Use Natural Language Processing (NLP) and Machine Learning (ML) to understand user intent
- **Strengths:** Adaptable, can learn over time, handle complex conversations, provide personalized responses.
- **Weaknesses:** More expensive and time-consuming to develop, requires a lot of data for training.



Key Social Media Platforms for Chatbots

- **Facebook Messenger:** The original pioneer for social media chatbots. Over 300,000 bots are currently active.
- **Instagram:** Focuses on visual commerce. Bots can respond to comments, story mentions, and DMs to drive engagement and sales.
- **WhatsApp:** A massive private messaging platform, ideal for customer service and high-touch communication. Businesses use it for order confirmations, shipping updates, and one-on-one support
- **Others:** Telegram, Slack, and even X (formerly Twitter) DMs are also used.



Use Cases: Customer Service

- **24/7 Availability:** Bots can answer questions and resolve issues outside of business hours.
- **First-line Support:** Triage customer queries, answer FAQs, and escalate complex issues to a human agent.
- **Issue Resolution:** Automate common tasks like checking order status, tracking shipments, or processing returns.
- **Example:** A chatbot for an e-commerce store can instantly provide a customer's order details without needing a human.



Use Cases: Marketing & Sales

- **Lead Generation:** Qualify leads by asking a series of questions and collecting user information.
- **Product Discovery:** Guide users through a product catalog based on their preferences.
- **Promotions:** Send out personalized offers, coupons, and news to segmented audiences.
- **Example:** Domino's Pizza's chatbot allows users to order a pizza directly from Facebook Messenger. This is a classic example of a transactional bot that drives sales.



Use Cases: Entertainment & Engagement

- **Interactive Stories:** Create engaging, multi-chapter stories with a "choose your own adventure" style.
- **Quizzes & Games:** Offer interactive quizzes and casual games to keep users on the platform and increase brand recall.
- **Content Curation:** Provide personalized news, articles, or fun facts based on user interests.
- **Example:** A news organization could have a chatbot that sends users a daily news digest on their chosen topics.



The Technology Stack

- **Natural Language Processing (NLP):** The engine that allows the chatbot to understand human language, including syntax, semantics, and intent.
- **Machine Learning (ML):** Used to train the bot to improve its responses over time by analyzing past conversations.
- **Platform APIs:** The interfaces that allow the chatbot to connect to social media platforms (e.g., Facebook Messenger Platform API).
- **Back-end System:** Manages the conversation flow, retrieves information from databases, and connects to other services (e.g., a CRM or e-commerce platform).



Building a Chatbot: The Process

1. **Define the Goal:** What specific problem will the bot solve? (e.g., reduce customer support calls, increase sales).
2. **Design the Conversation Flow:** Map out the user journey and potential conversation paths.
3. **Choose the Right Tools:** Select a platform (e.g., Chatfuel, ManyChat) or a custom development approach.
4. **Train the Bot:** Feed the bot with sample conversations and data to improve its understanding.
5. **Test and Deploy:** Test the bot with real users to identify and fix issues.
6. **Analyze and Iterate:** Continuously monitor performance metrics and make improvements.





Best Practices for Success

- **Give it a Persona:** Design a unique personality for your bot that aligns with your brand. Is it friendly, formal, or witty?
- **Set Clear Expectations:** Let the user know they are talking to a bot and what it can and cannot do.
- **Seamless Handover:** Always provide an easy and clear path for the user to connect with a human agent.
- **Handle Errors Gracefully:** When the bot doesn't understand, provide helpful, polite, and actionable responses. Avoid frustration.



Challenges & Limitations

- **Misunderstanding Intent:** The bot may misinterpret complex or nuanced human language, leading to a poor user experience.
- **Privacy & Data Security:** Handling sensitive user information requires robust security measures and clear privacy policies.
- **Maintenance:** Bots require ongoing training and maintenance to remain relevant and effective.
- **User Frustration:** A poorly designed bot can be more frustrating than having no bot at all.



The Future of Conversational Interfaces

- **Multimodal AI:** Chatbots that can understand and respond with not just text, but also images, video, and audio.
- **Voice Commerce:** More brands will adopt voice-based interfaces for transactions and support.
- **Personalization at Scale:** Generative AI will allow for hyper-personalized, dynamic, and fluid conversations.
- **The Metaverse:** Conversational interfaces will be essential for navigating and interacting in virtual and augmented reality environments.



THANK YOU !